

Wei Chen

chenwei8994@126.com | github.com/CWhhhaaha
+86 19980818994 | International: +1 6085093756

Education

University of Wisconsin–Madison

Aug 2024 – Dec 2025

M.S. in Electrical and Computer Engineering

Machine Learning and Signal Processing (MLSP), GPA: 3.73 / 4.00

Sichuan University

Sep 2020 – Jun 2024

B.S. in Flight Vehicle Control and Information Engineering

GPA: 3.79 / 4.00

Selected Projects

My Transformer

[GitHub](#)

- Built a from-scratch PyTorch encoder-decoder Transformer for machine translation, including multi-head attention, positional encoding, training, inference, and evaluation.
- Implemented data preparation, greedy / beam-search decoding, BLEU evaluation, checkpointing, testing, and attention visualization.

Global-Market-Newsboard

[GitHub](#)

- Developed an open-source market news dashboard focused on turning real-time information streams into a more usable interface for retail traders.
- Built a user-facing workflow spanning news aggregation, data handling, presentation logic, and practical decision-support design; deployed at globalnewsboard.cn (demo password: 123456).

Campus Lane Detection

[GitHub](#)

- Built a course project around ultra-fast deep lane detection in PyTorch using a structure-aware lane detection pipeline with training, evaluation, visualization, and speed-testing scripts.
- Adapted TuSimple / CULane-style tooling to campus driving footage and produced a 1080p / 30fps curved-lane overlay demo.

Research Manuscripts

- *Basis-Mediated Bilinear Attention: A New Method for Greatly Reducing Query–Key Pathway Parameters*. Submitted to NeurIPS 2026. Proposes BMB/BMB-UV for query-key reparameterization in 7B-scale pretraining, reducing QK parameters by up to 95.83% while preserving model quality.
- *Revisiting Positional Priors in Vision Transformers with Semantic Gating*. Manuscript in preparation on semantic gating beyond static geometry-driven bias in ViT attention.
- *Beyond Positional Encoding: Layerwise Attention Reconstruction via Query-Key Structural Priors*. Manuscript in preparation on layerwise attention reconstruction using query-key structural priors.

Skills

- **LLM Pretraining**: distributed training, data-parallel training, mixed precision, checkpointing, throughput profiling, training stability
- **Memory and Efficiency**: activation checkpointing, FlashAttention, KV-cache optimization, quantization, parameter-efficient tuning
- **Optimization and Scaling**: Adam/AdamW, Lion, Shampoo, SOAP, Muon; scaling laws, data recipes, compute-data trade-offs, model-size planning
- **Post-Training and Adaptation**: SFT, RLHF, preference optimization, LoRA, DoRA, QLoRA, instruction-data refinement
- **Agentic Workflows**: tool use, ReAct-style reasoning, multi-agent coordination, prompt/program optimization; Codex, Claude Code, Kimi, Swarm

Selected Coursework

- Large Language Models and Deep Learning: [Advanced Topics in Large Language Models \(ECE 901\)](#); [Introduction to Artificial Neural Networks \(CS 539\)](#)
- Machine Learning and Optimization: [Machine Learning \(ECE 760\)](#); [Matrix Methods in Machine Learning \(ECE 532\)](#); [Topics in Mathematical Data Science \(ECE 888\)](#)
- Perception and Signal Processing: [Image Processing \(ECE 533\)](#)